ASignX Normalized Data Dictionary

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This dictionary defines the 16 normalized application tables used by ASignX for ASign Learn and ASignLive. It lists all 89 application columns, their PostgreSQL types, required values, defaults, keys, relationships, and validation rules.

Supabase Auth owns auth.users. A separate reference section documents the user attributes relevant to ASignX. It is not a seventeenth application table, and the application migrations do not recreate it.

The tables separate account profiles, reusable learning content, enrollment activity, recognition outcomes, and achievement definitions. Junction tables represent many-to-many relationships. Total XP and streaks are derived from activity instead of being duplicated as profile balances.

Reading the dictionary

PK means primary key; FK means foreign key. Multiple PK fields in one table form one composite primary key. A composite FK must be satisfied by the field combination, not by each field independently.

None means there is no database column default. The caller must provide a required value unless a documented trigger derives it. NULL means the field is optional and has no explicit default. Managed means Supabase controls the external field default.

VARCHAR sizes are character limits. NUMERIC(6,5) has six total digits and five fractional digits; score checks limit values to 0 through 1. TIMESTAMPTZ represents an instant; profile timezone is used separately for local learning dates.

CASCADE removes dependent records with their parent. RESTRICT prevents deleting a referenced shared record. All application FK updates use NO ACTION. RLS means row-level security.

Supabase Auth users reference

Selected Supabase-managed account attributes. These physical fields are reference information, not application-owned DDL. Supabase controls additional columns, indexes, and defaults; application relationships reference only its stable id primary key.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
id	UUID	128 bits	Yes	-	No	Managed	Authentication identity referenced by profiles.id.
email	VARCHAR	255 characters	-	-	Yes	Managed	Email used for email-based sign-in. Not duplicated in profiles.
encrypted_password	VARCHAR	255 characters	-	-	Yes	Managed	Secure password hash for password-based accounts; not a readable password.
last_sign_in_at	TIMESTAMPTZ	Microseconds	-	-	Yes	Managed	Most recent sign-in time, maintained by Supabase Auth.
raw_user_meta_data	JSONB	Variable	-	-	Yes	Managed	User metadata; initial display name can seed profiles. Never trusted for assigning application roles.
created_at	TIMESTAMPTZ	Microseconds	-	-	Yes	Managed	Authentication record creation time.
updated_at	TIMESTAMPTZ	Microseconds	-	-	Yes	Managed	Authentication record modification time.

- One Auth identity has at most one application profile by key structure. Signup and backfill triggers create profiles with the same UUID.
- The encrypted_password field contains a password hash, despite its historical column name. Applications use Supabase Auth for password operations and do not copy it into profiles.
- Nullable email and password fields accommodate account types beyond email/password. ASignX email/password registration still requires those inputs through Auth.

1 profiles

Application information for a registered learner or administrator. Authentication stays in Supabase Auth.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
id	UUID	128 bits	Yes	auth.users.id	No	None	Same UUID as auth.users.id; supplied by the Auth profile-creation trigger, with no column default.
display_name	VARCHAR	100 characters	-	-	No	None	Name displayed in the application.
role	VARCHAR	10 characters	-	-	No	'learner'	ASignX authorization role: learner or admin. Separate from Supabase Auth database roles.
timezone	VARCHAR	64 characters	-	-	No	'UTC'	Valid IANA time zone used when deriving learning streaks.
created_at	TIMESTAMPTZ	Microseconds	-	-	No	now()	Profile creation time.
updated_at	TIMESTAMPTZ	Microseconds	-	-	No	now()	Profile modification time; refreshed by the validation trigger on UPDATE.

Keys and constraints

- Primary key: (id).
- Check: length(btrim(display_name)) > 0.
- Check: role in ('learner', 'admin').
- Foreign key: (id) -> auth.users(id); ON DELETE CASCADE; ON UPDATE NO ACTION.

Validation and access rules

- Database trigger validates timezone and refreshes updated_at on profile updates.
- An Auth signup trigger creates a learner profile. Existing Auth users are backfilled. The display name is taken from display_name or name metadata, falling back to ASL explorer; the application role is never copied from signup metadata.
- A signed-in learner may read their own profile and update only display_name and timezone. Role changes require the trusted backend.

2 signs

Shared ASL vocabulary for demonstrations, camera practice, and Live recognition.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
id	UUID	128 bits	Yes	-	No	gen_random_uuid()	Unique sign identifier.
gloss	VARCHAR	100 characters	-	-	No	None	Unique vocabulary label, such as HELLO or THANK-YOU.
meaning	TEXT	Variable	-	-	No	None	Meaning of the sign.
category	VARCHAR	60 characters	-	-	Yes	NULL	Simple browsing label, such as Greetings or Numbers; no separate taxonomy is needed.
instructions	TEXT	Variable	-	-	Yes	NULL	Optional guidance for performing the sign.
demo_path	TEXT	Variable	-	-	Yes	NULL	Object path for one curated demonstration video in the learning-media Storage bucket.
is_published	BOOLEAN	1 byte	-	-	No	false	Whether learners may access this sign.
created_at	TIMESTAMPTZ	Microseconds	-	-	No	now()	Sign creation time.

Keys and constraints

- Primary key: (id).
- Unique: (gloss).
- Check: length(btrim(gloss)) > 0.
- Check: length(btrim(meaning)) > 0.

Validation and access rules

- Public readers see only published signs. demo_path is a Storage object path for curated learning media.

3 courses

Learning courses available for enrollment.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
id	UUID	128 bits	Yes	-	No	gen_random_uuid()	Unique course identifier.
title	VARCHAR	150 characters	-	-	No	None	Course title.
description	TEXT	Variable	-	-	Yes	NULL	Course overview.
is_published	BOOLEAN	1 byte	-	-	No	false	Whether the course is available to learners.
created_at	TIMESTAMPTZ	Microseconds	-	-	No	now()	Course creation time.

Keys and constraints

- Primary key: (id).
- Check: length(btrim(title)) > 0.

Validation and access rules

- Public readers see only published courses. A course can have no lessons or enrollments when first created.

4 lessons

Ordered learning units in a course. Learning text and linked sign demonstrations supply the lesson material.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
id	UUID	128 bits	Yes	-	No	gen_random_uuid()	Unique lesson identifier.
course_id	UUID	128 bits	-	courses.id	No	None	References courses.id.
title	VARCHAR	150 characters	-	-	No	None	Lesson title.
content	TEXT	Variable	-	-	Yes	NULL	Instructional text; demonstrations are reached through lesson_signs.
sort_order	INTEGER	32 bits	-	-	No	None	Positive lesson position within its course.
xp_reward	INTEGER	32 bits	-	-	No	10	Nonnegative XP awarded once on validated lesson completion.
is_published	BOOLEAN	1 byte	-	-	No	false	Whether this lesson is visible within its published course.

Keys and constraints

- Primary key: (id).
- Unique: (course_id, sort_order).
- Check: length(btrim(title)) > 0.
- Check: sort_order > 0.
- Check: xp_reward >= 0.
- Foreign key: (course_id) -> public.courses(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.

Validation and access rules

- Public readers see a lesson only when both the lesson and its parent course are published.
- A deferred publication trigger requires at least two choices and exactly one correct choice for each exercise. A lesson may have no exercises.
- The lesson identity and parent course cannot be changed by UPDATE. The configured XP is captured when the lesson is first completed.

5 lesson signs

Connects signs to lessons so the same sign can appear in several lessons.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
lesson_id	UUID	128 bits	Yes	lessons.id	No	None	References lessons.id; part of the composite primary key.
sign_id	UUID	128 bits	Yes	signs.id	No	None	References signs.id; part of the composite primary key.
sort_order	INTEGER	32 bits	-	-	No	None	Positive demonstration order within the lesson.

Keys and constraints

- Primary key: (lesson_id, sign_id).
- Unique: (lesson_id, sort_order).
- Check: sort_order > 0.
- Foreign key: (lesson_id) -> public.lessons(id); ON DELETE CASCADE; ON UPDATE NO ACTION.
- Foreign key: (sign_id) -> public.signs(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.

Validation and access rules

- The pair (lesson_id, sign_id) represents a many-to-many link. Public readers require a published lesson, parent course, and linked sign.

6 exercises

Each exercise is one multiple-choice question in a lesson. Several exercises together form a quiz; camera practice is recorded separately.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
id	UUID	128 bits	Yes	-	No	gen_random_uuid()	Unique question/exercise identifier.
lesson_id	UUID	128 bits	-	lessons.id	No	None	References lessons.id.
prompt	TEXT	Variable	-	-	No	None	Question displayed to the learner.
sign_id	UUID	128 bits	-	signs.id	Yes	NULL	Optional reference to signs.id for a sign-based question.
sort_order	INTEGER	32 bits	-	-	No	None	Positive exercise position within its lesson.

Keys and constraints

- Primary key: (id).
- Unique: (lesson_id, sort_order).
- Check: length(btrim(prompt)) > 0.
- Check: sort_order > 0.
- Foreign key: (lesson_id) -> public.lessons(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.
- Foreign key: (sign_id) -> public.signs(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.

Validation and access rules

- Each row is one multiple-choice question; the collection of questions in a lesson forms its quiz.
- The parent lesson cannot be reassigned. Definition edits are rejected while the lesson is published, after the exercise has attempts, or after its lesson has completed progress. Create a new content version for substantive changes.

7 exercise options

Answer choices for an exercise. Correctness is server-only and must not be included in learner responses.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
id	UUID	128 bits	Yes	-	No	gen_random_uuid()	Unique option identifier.
exercise_id	UUID	128 bits	-	exercises.id	No	None	References exercises.id.
option_text	TEXT	Variable	-	-	No	None	Answer choice text displayed to learners.
is_correct	BOOLEAN	1 byte	-	-	No	false	Marks the correct option; read only by the trusted grading/authoring backend.
sort_order	INTEGER	32 bits	-	-	No	None	Positive option order within the exercise.

Keys and constraints

- Primary key: (id).
- Unique: (exercise_id, sort_order).
- Unique: (exercise_id, id).
- Check: length(btrim(option_text)) > 0.
- Check: sort_order > 0.
- Foreign key: (exercise_id) -> public.exercises(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.

Validation and access rules

- The partial unique index on exercise_id WHERE is_correct allows at most one correct option per exercise; lesson publication requires exactly one.
- Direct browser reads are denied. The backend exposes only id, exercise_id, option_text, and sort_order, excluding is_correct.
- Options follow the same published/used definition freeze as their exercise, and cannot be moved to another exercise.

8 enrollments

Connects a learner to a course and provides the parent for learning activity in that course.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
id	UUID	128 bits	Yes	-	No	gen_random_uuid()	Unique enrollment identifier.
user_id	UUID	128 bits	-	profiles.id	No	None	References profiles.id.
course_id	UUID	128 bits	-	courses.id	No	None	References courses.id.
enrolled_at	TIMESTAMPTZ	Microseconds	-	-	No	now()	Enrollment time.

Keys and constraints

- Primary key: (id).
- Unique: (user_id, course_id).
- Foreign key: (user_id) -> public.profiles(id); ON DELETE CASCADE; ON UPDATE NO ACTION.
- Foreign key: (course_id) -> public.courses(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.

Validation and access rules

- One enrollment per learner and course. Learning activity reaches its owner through this row.
- Enrollment identity, learner, course, and enrollment time are immutable. Browser clients may read their own enrollments; creation is a backend operation.

9 lesson progress

Tracks each lesson in an enrollment and stores its one-time completion reward; a separate XP ledger is unnecessary for this reward policy.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
enrollment_id	UUID	128 bits	Yes	enrollments.id	No	None	References enrollments.id; part of the composite primary key.
lesson_id	UUID	128 bits	Yes	lessons.id	No	None	References lessons.id; part of the composite primary key.
started_at	TIMESTAMPTZ	Microseconds	-	-	No	now()	Time the learner started the lesson.
completed_at	TIMESTAMPTZ	Microseconds	-	-	Yes	NULL	Validated completion time; NULL means in progress.
xp_earned	INTEGER	32 bits	-	-	No	0	Nonnegative first-completion XP actually granted; remains zero until completion.

Keys and constraints

- Primary key: (enrollment_id, lesson_id).
- Check: xp_earned >= 0.
- Check: (completed_at is null and xp_earned = 0) or (completed_at is not null and completed_at >= started_at).
- Foreign key: (enrollment_id) -> public.enrollments(id); ON DELETE CASCADE; ON UPDATE NO ACTION.
- Foreign key: (lesson_id) -> public.lessons(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.

Validation and access rules

- A trigger verifies that the lesson belongs to the enrollment course. Progress identity and started_at are immutable.
- Before completion, xp_earned is forced to zero. Completion requires at least one correct attempt for every exercise and replaces supplied XP with lessons.xp_reward.
- Completed timestamps and XP are immutable. Completed progress cannot be independently deleted while its enrollment remains. Account/enrollment deletion may cascade.
- For a lesson without exercises, the backend must confirm that its material was viewed before requesting completion; the database does not observe viewing.

10 exercise attempts

A learner submission for one multiple-choice exercise. Multiple attempts are permitted.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
id	UUID	128 bits	Yes	-	No	gen_random_uuid()	Unique attempt identifier; reuse this ID for a retried submission.
enrollment_id	UUID	128 bits	-	enrollments.id	No	None	References enrollments.id; identifies the learner through enrollment.
exercise_id	UUID	128 bits	-	exercises.id; exercise_options.exercise_id (composite)	No	None	References exercises.id.
selected_option_id	UUID	128 bits	-	exercise_options.id (composite)	No	None	References exercise_options.id; must belong to exercise_id using the composite foreign key below.
is_correct	BOOLEAN	1 byte	-	-	No	None	Database-trigger grading result, derived from the selected option on insert.
attempted_at	TIMESTAMPTZ	Microseconds	-	-	No	now()	Submission time.

Keys and constraints

- Primary key: (id).
- Foreign key: (enrollment_id) -> public.enrollments(id); ON DELETE CASCADE; ON UPDATE NO ACTION.
- Foreign key: (exercise_id) -> public.exercises(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.
- Foreign key: (exercise_id, selected_option_id) -> public.exercise_options(exercise_id, id); ON DELETE RESTRICT; ON UPDATE NO ACTION.

Validation and access rules

- The composite foreign key (exercise_id, selected_option_id) references exercise_options(exercise_id, id), ensuring the selected option belongs to this exercise.
- A trigger verifies course membership and derives is_correct from the stored answer option. Caller-supplied correctness is overwritten.
- Attempt rows cannot be updated. Reuse the submission UUID with conflict handling for a retried request.

11 model versions

Minimal record of the actual ML model used for a prediction; deployment and evaluation tooling remain outside this application schema.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
id	UUID	128 bits	Yes	-	No	gen_random_uuid()	Unique model version identifier.
version	VARCHAR	50 characters	-	-	No	None	Unique immutable release name.
artifact_path	TEXT	Variable	-	-	No	None	Private object path in the model-artifacts Storage bucket.
confidence_threshold	NUMERIC	6 digits total, 5 decimal	-	-	No	0.80000	Acceptance threshold from 0 to 1 for this model version.
created_at	TIMESTAMPTZ	Microseconds	-	-	No	now()	Model registration time.

Keys and constraints

- Primary key: (id).
- Unique: (version).
- Check: length(btrim(version)) > 0.
- Check: length(btrim(artifact_path)) > 0.
- Check: confidence_threshold between 0 and 1.

Validation and access rules

- All model-version updates are rejected. Register a new row when the version, artifact, or confidence threshold changes.
- artifact_path refers to a private model-artifacts Storage object. This table has no direct browser access.

12 practice attempts

Authenticated camera practice results, including standalone practice and optional lesson practice.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
id	UUID	128 bits	Yes	-	No	gen_random_uuid()	Unique practice attempt identifier; reuse on retry.
user_id	UUID	128 bits	-	profiles.id	No	None	References profiles.id.
lesson_id	UUID	128 bits	-	lessons.id	Yes	NULL	Optional references lessons.id; NULL means standalone practice or review.
target_sign_id	UUID	128 bits	-	signs.id	No	None	References signs.id; the sign the learner was asked to perform.
predicted_sign_id	UUID	128 bits	-	signs.id	Yes	NULL	References signs.id; NULL means no supported sign was recognized.
model_version_id	UUID	128 bits	-	model_versions.id	No	None	References model_versions.id; actual model used by the recognition service.
confidence	NUMERIC	6 digits total, 5 decimal	-	-	No	None	Recognition score between 0 and 1.
is_correct	BOOLEAN	1 byte	-	-	No	None	Database-trigger result based on target match and the recorded model confidence threshold.
attempted_at	TIMESTAMPTZ	Microseconds	-	-	No	now()	Practice time.

Keys and constraints

- Primary key: (id).
- Check: confidence between 0 and 1.
- Foreign key: (user_id) -> public.profiles(id); ON DELETE CASCADE; ON UPDATE NO ACTION.
- Foreign key: (lesson_id) -> public.lessons(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.
- Foreign key: (target_sign_id) -> public.signs(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.
- Foreign key: (predicted_sign_id) -> public.signs(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.
- Foreign key: (model_version_id) -> public.model_versions(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.

Validation and access rules

- For lesson practice, a trigger requires enrollment in the lesson course and target membership in lesson_signs. Standalone practice has no lesson_id.
- The database derives is_correct as prediction equals target and confidence meets the referenced model threshold; a missing prediction yields false.
- Updates are rejected. The backend supplies a trusted recognition result; storing it does not itself implement model inference.

13 recognition sessions

Optional saved ASignLive history for authenticated users. Guest and unsaved sessions are transient.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
id	UUID	128 bits	Yes	-	No	gen_random_uuid()	Unique saved session identifier.
user_id	UUID	128 bits	-	profiles.id	No	None	References profiles.id; session owner.
started_at	TIMESTAMPTZ	Microseconds	-	-	No	now()	Session start time.
ended_at	TIMESTAMPTZ	Microseconds	-	-	Yes	NULL	Session end time; NULL while open.
final_text	TEXT	Variable	-	-	Yes	NULL	Text assembled by the finalization trigger from accepted sign meanings; NULL while open.

Keys and constraints

- Primary key: (id).
- Check: ended_at is null or ended_at >= started_at.
- Check: final_text is null or ended_at is not null.
- Foreign key: (user_id) -> public.profiles(id); ON DELETE CASCADE; ON UPDATE NO ACTION.

Validation and access rules

- The backend creates a saved session only after explicit opt-in. Consent and authentication are application responsibilities; there is no consent column in this schema.
- Guest and unsaved sessions are transient. Only the owner may read saved history.
- When ended, a trigger composes final_text from sign meanings in sequence_no order, accepting only known signs meeting each result model threshold. With no accepted result, final_text is an empty string.
- The session identity, owner, and started_at are immutable; an ended session cannot be updated.

14 recognition results

Ordered stabilized sign predictions within a saved Live session; rows themselves represent the recognized sign sequence.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
id	UUID	128 bits	Yes	-	No	gen_random_uuid()	Unique prediction-event identifier.
session_id	UUID	128 bits	-	recognition_sessions.id	No	None	References recognition_sessions.id.
sign_id	UUID	128 bits	-	signs.id	Yes	NULL	References signs.id; NULL means no supported label was recognized.
model_version_id	UUID	128 bits	-	model_versions.id	No	None	References model_versions.id; actual model used for this event.
sequence_no	INTEGER	32 bits	-	-	No	None	Positive event position within the session.
confidence	NUMERIC	6 digits total, 5 decimal	-	-	No	None	Recognition score between 0 and 1; only accepted labeled results contribute to final_text.
recognized_at	TIMESTAMPTZ	Microseconds	-	-	No	now()	Prediction time.

Keys and constraints

- Primary key: (id).
- Unique: (session_id, sequence_no).
- Check: sequence_no > 0.
- Check: confidence between 0 and 1.
- Foreign key: (session_id) -> public.recognition_sessions(id); ON DELETE CASCADE; ON UPDATE NO ACTION.
- Foreign key: (sign_id) -> public.signs(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.
- Foreign key: (model_version_id) -> public.model_versions(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.

Validation and access rules

- The database rejects inserts into ended sessions and prevents UPDATE of stored results.
- sequence_no determines ordering. Unknown or low-confidence results remain recorded but do not contribute to final_text.
- The backend persists stabilized sign events, not camera frames. Only the owner of the parent session may read these rows.

15 achievements

Simple badge definitions based on a supported learning metric and threshold.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
id	UUID	128 bits	Yes	-	No	gen_random_uuid()	Unique achievement identifier.
code	VARCHAR	50 characters	-	-	No	None	Unique stable badge code, such as FIRST_LESSON.
name	VARCHAR	100 characters	-	-	No	None	Badge name displayed to learners.
description	TEXT	Variable	-	-	No	None	Description of how to earn the badge.
metric	VARCHAR	25 characters	-	-	No	None	One of completed_lessons, successful_practices, total_xp, or streak_days.
threshold	INTEGER	32 bits	-	-	No	None	Positive target value for the selected metric.

Keys and constraints

- Primary key: (id).
- Unique: (code).
- Check: length(btrim(code)) > 0.
- Check: length(btrim(name)) > 0.
- Check: length(btrim(description)) > 0.
- Check: metric in ('completed_lessons', 'successful_practices', 'total_xp', 'streak_days').
- Check: threshold > 0.

Validation and access rules

- Supported metrics are completed_lessons, successful_practices, total_xp, and streak_days.
- Once any user has received the achievement, its definition cannot be updated. Use a new achievement for revised criteria.

16 user achievements

Badges unlocked by learners; prevents duplicate awards.

Column name	Data type	Size or precision	PK	FK reference	Nullable	Default value	Description
user_id	UUID	128 bits	Yes	profiles.id	No	None	References profiles.id; part of the composite primary key.
achievement_id	UUID	128 bits	Yes	achievements.id	No	None	References achievements.id; part of the composite primary key.
unlocked_at	TIMESTAMPTZ	Microseconds	-	-	No	now()	Time the backend confirmed eligibility and awarded the badge.

Keys and constraints

- Primary key: (user_id, achievement_id).
- Foreign key: (user_id) -> public.profiles(id); ON DELETE CASCADE; ON UPDATE NO ACTION.
- Foreign key: (achievement_id) -> public.achievements(id); ON DELETE RESTRICT; ON UPDATE NO ACTION.

Validation and access rules

- A trigger validates the referenced achievement metric and threshold before an award is inserted. The composite primary key prevents duplicate awards.
- Award records cannot be updated. Learners may read only their own awards.

Relationship reference

Parent tables may have zero or many children. Required or optional describes the child foreign key. The profile relationship allows at most one profile per Auth identity.

Parent table	Child table and FK fields	Parent to child	Child reference	On delete
auth.users	profiles(id)	One to zero or one	Required	CASCADE
public.courses	lessons(course_id)	One to many	Required	RESTRICT
public.lessons	lesson_signs(lesson_id)	One to many	Required	CASCADE
public.signs	lesson_signs(sign_id)	One to many	Required	RESTRICT
public.lessons	exercises(lesson_id)	One to many	Required	RESTRICT
public.signs	exercises(sign_id)	One to many	Optional	RESTRICT
public.exercises	exercise_options(exercise_id)	One to many	Required	RESTRICT
public.profiles	enrollments(user_id)	One to many	Required	CASCADE
public.courses	enrollments(course_id)	One to many	Required	RESTRICT
public.enrollments	lesson_progress(enrollment_id)	One to many	Required	CASCADE
public.lessons	lesson_progress(lesson_id)	One to many	Required	RESTRICT
public.enrollments	exercise_attempts(enrollment_id)	One to many	Required	CASCADE
public.exercises	exercise_attempts(exercise_id)	One to many	Required	RESTRICT
public.exercise_options	exercise_attempts(exercise_id, selected_option_id)	One to many	Required	RESTRICT
public.profiles	practice_attempts(user_id)	One to many	Required	CASCADE
public.lessons	practice_attempts(lesson_id)	One to many	Optional	RESTRICT
public.signs	practice_attempts(target_sign_id)	One to many	Required	RESTRICT
public.signs	practice_attempts(predicted_sign_id)	One to many	Optional	RESTRICT
public.model_versions	practice_attempts(model_version_id)	One to many	Required	RESTRICT
public.profiles	recognition_sessions(user_id)	One to many	Required	CASCADE
public.recognition_sessions	recognition_results(session_id)	One to many	Required	CASCADE
public.signs	recognition_results(sign_id)	One to many	Optional	RESTRICT
public.model_versions	recognition_results(model_version_id)	One to many	Required	RESTRICT

Parent table	Child table and FK fields	Parent to child	Child reference	On delete
public.profiles	user_achievements(user_id)	One to many	Required	CASCADE
public.achievements	user_achievements(achievement_id)	One to many	Required	RESTRICT

Shared rules

- All 16 application tables enable RLS. Public roles can read published learning content and achievement definitions. A lesson also requires a published parent course, and a lesson-sign link additionally requires a published sign.
- Authenticated browser users can read their own profile, enrollments, progress, attempts, practices, saved sessions and results, and earned achievements. The only browser profile writes are display_name and timezone. The backend validates caller identity, ownership, and administrator authorization for privileged operations.
- exercise_options and model_versions have no direct browser access. Safe answer fields are delivered through the backend; correctness and private model paths remain server-only.
- Guests may browse learning content and use transient practice and Live features without an application profile. Saving personal activity requires a registered account. Guest browser activity is not a trusted source of account XP, grading results, or awards.
- Total XP is SUM(lesson_progress.xp_earned) through the learner enrollments. Current streak uses distinct local dates with a completed lesson or successful practice; the consecutive run must end today or yesterday in profiles.timezone. A timezone change can change the derived date boundaries.
- Deleting an Auth user cascades to the profile and its dependent personal activity. Shared vocabulary, courses, lessons, models, and achievement definitions remain; referenced shared content should be unpublished or archived rather than removed.
- Saving Live history requires explicit user opt-in in the application. The schema stores stabilized events and finalized text, with no camera-frame or raw-video columns. Curated media and model files belong in Storage; database fields contain object paths.
- Backend services must verify viewing for no-quiz completion, validate recognition service output, enforce opt-in, and expose only authorized fields. Database constraints and triggers do not replace these application responsibilities.

Normalization notes

- Identity and profile: authentication credentials remain in the managed Auth schema; application attributes depend on the profile UUID. Repeating email and password columns in application activity tables are avoided.
- Reusable content: signs, courses, lessons, exercises, and answer options have separate identities. lesson_signs resolves the many-to-many association between lessons and signs.
- Learner activity: enrollments resolves learners and courses; lesson_progress depends on an enrollment and lesson; exercise_attempts records individual submissions rather than repeating course or profile details.
- Prediction provenance: model_versions stores model artifact and threshold definitions once. Practice and recognition rows reference the actual model version used.
- Achievement definitions and awards: achievements defines criteria once; user_achievements resolves the many-to-many association and prevents duplicate awards.
- Historical snapshots: xp_earned and is_correct are intentional recorded outcomes. They preserve the result of an event rather than copying mutable profile totals. Unique order constraints and the option-pair foreign key enforce additional candidate keys and consistency.

References

Application schema: server/src/db/schema/app.ts

Generated table and constraint migration: server/drizzle/20261021161839_asignx_application/migration.sql

Custom validation triggers and access grants: server/drizzle/20261021161905_asignx_rules/migration.sql

Supabase account profile guidance: https://supabase.com/docs/guides/auth/managing-user-data

Supabase password storage guidance: https://supabase.com/docs/guides/auth/password-security

Supabase Auth user field definitions: https://github.com/supabase/auth/blob/master/migrations/00_init_auth_schema.up.sql